

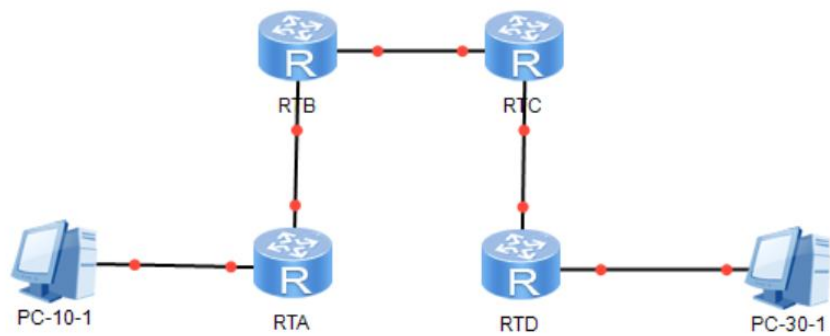
Lab-4 Static and Default Routing

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一、Lab-4-1: Static and Default Route Configuration

Q4-1.1~4-1.8:

(1) 创建的拓补图如下:



(2) RTA 的 IP 路由表如下:

```
[RTA]display IP routing-table
Route Flags: R - relay, D - download to fib
-----
Routing Tables: Public
  Destinations : 10      Routes : 10

Destination/Mask    Proto   Pre  Cost   Flags NextHop         Interface
-----
 127.0.0.0/8        Direct  0    0       D  127.0.0.1       InLoopBack0
 127.0.0.1/32        Direct  0    0       D  127.0.0.1       InLoopBack0
127.255.255.255/32   Direct  0    0       D  127.0.0.1       InLoopBack0
172.16.101.0/24      Direct  0    0       D  172.16.101.1    GigabitEthernet
0/0/0
172.16.101.1/32      Direct  0    0       D  127.0.0.1       GigabitEthernet
0/0/0
172.16.101.255/32    Direct  0    0       D  127.0.0.1       GigabitEthernet
0/0/0
192.168.10.0/24      Direct  0    0       D  192.168.10.1    GigabitEthernet
0/0/2
192.168.10.1/32      Direct  0    0       D  127.0.0.1       GigabitEthernet
0/0/2
192.168.10.255/32    Direct  0    0       D  127.0.0.1       GigabitEthernet
0/0/2
255.255.255.255/32   Direct  0    0       D  127.0.0.1       InLoopBack0
```

(3) RTB 的 IP 路由表如下:

```
[RTB]display IP routing-table
Route Flags: R - relay, D - download to fib
-----
Routing Tables: Public
  Destinations : 10      Routes : 10

Destination/Mask    Proto   Pre  Cost   Flags NextHop         Interface
-----
 127.0.0.0/8        Direct  0    0       D  127.0.0.1       InLoopBack0
 127.0.0.1/32        Direct  0    0       D  127.0.0.1       InLoopBack0
127.255.255.255/32   Direct  0    0       D  127.0.0.1       InLoopBack0
172.16.101.0/24      Direct  0    0       D  172.16.101.2    GigabitEthernet
0/0/0
172.16.101.2/32      Direct  0    0       D  127.0.0.1       GigabitEthernet
0/0/0
172.16.101.255/32    Direct  0    0       D  127.0.0.1       GigabitEthernet
0/0/0
172.16.102.0/24      Direct  0    0       D  172.16.102.1    GigabitEthernet
0/0/1
172.16.102.1/32      Direct  0    0       D  127.0.0.1       GigabitEthernet
0/0/1
172.16.102.255/32    Direct  0    0       D  127.0.0.1       GigabitEthernet
0/0/1
255.255.255.255/32   Direct  0    0       D  127.0.0.1       InLoopBack0
```

(4) RTC 的 IP 路由表如下:

```
[RTC]display IP routing-table
Route Flags: R - relay, D - download to fib
-----
Routing Tables: Public
  Destinations : 10          Routes : 10

Destination/Mask    Proto   Pre  Cost   Flags NextHop         Interface
-----
127.0.0.0/8         Direct  0    0       D  127.0.0.1       InLoopBack0
127.0.0.1/32        Direct  0    0       D  127.0.0.1       InLoopBack0
127.255.255.255/32   Direct  0    0       D  127.0.0.1       InLoopBack0
172.16.102.0/24      Direct  0    0       D  172.16.102.2    GigabitEthernet
0/0/1
172.16.102.2/32      Direct  0    0       D  127.0.0.1       GigabitEthernet
0/0/1
172.16.102.255/32    Direct  0    0       D  127.0.0.1       GigabitEthernet
0/0/1
172.16.103.0/24      Direct  0    0       D  172.16.103.2    GigabitEthernet
0/0/0
172.16.103.2/32      Direct  0    0       D  127.0.0.1       GigabitEthernet
0/0/0
172.16.103.255/32    Direct  0    0       D  127.0.0.1       GigabitEthernet
0/0/0
255.255.255.255/32   Direct  0    0       D  127.0.0.1       InLoopBack0
```

(5) RTD 的 IP 路由表如下:

```
[RTD]display IP routing-table
Route Flags: R - relay, D - download to fib
-----
Routing Tables: Public
  Destinations : 13          Routes : 13

Destination/Mask    Proto   Pre  Cost   Flags NextHop         Interface
-----
127.0.0.0/8         Direct  0    0       D  127.0.0.1       InLoopBack0
127.0.0.1/32        Direct  0    0       D  127.0.0.1       InLoopBack0
127.255.255.255/32   Direct  0    0       D  127.0.0.1       InLoopBack0
172.16.103.0/24      Direct  0    0       D  172.16.103.1    GigabitEthernet
0/0/0
172.16.103.1/32      Direct  0    0       D  127.0.0.1       GigabitEthernet
0/0/0
172.16.103.255/32    Direct  0    0       D  127.0.0.1       GigabitEthernet
0/0/0
192.168.30.0/24      Direct  0    0       D  192.168.30.1    GigabitEthernet
0/0/2
192.168.30.1/32      Direct  0    0       D  127.0.0.1       GigabitEthernet
0/0/2
192.168.30.255/32    Direct  0    0       D  127.0.0.1       GigabitEthernet
0/0/2
192.168.200.0/24     Direct  0    0       D  192.168.200.100 LoopBack0
192.168.200.100/32   Direct  0    0       D  127.0.0.1       LoopBack0
192.168.200.255/32   Direct  0    0       D  127.0.0.1       LoopBack0
255.255.255.255/32   Direct  0    0       D  127.0.0.1       InLoopBack0
```

(6) PC-10-1 和 PC-30-1 不能 ping 通:

```
PC>ping 192.168.10.11

Ping 192.168.10.11: 32 data bytes, Press Ctrl_C to break
Request timeout!
Request timeout!
Request timeout!
Request timeout!
Request timeout!

--- 192.168.10.11 ping statistics ---
 5 packet(s) transmitted
 0 packet(s) received
100.00% packet loss
```

(7) PC-30-1 和 RTD 上的 loopback 0 可以 ping 通:

```

PC>ping 192.168.200.100

Ping 192.168.200.100: 32 data bytes, Press Ctrl_C to break
From 192.168.200.100: bytes=32 seq=1 ttl=255 time=15 ms
From 192.168.200.100: bytes=32 seq=2 ttl=255 time<1 ms
From 192.168.200.100: bytes=32 seq=3 ttl=255 time<1 ms
From 192.168.200.100: bytes=32 seq=4 ttl=255 time<1 ms
From 192.168.200.100: bytes=32 seq=5 ttl=255 time=16 ms

--- 192.168.200.100 ping statistics ---
 5 packet(s) transmitted
 5 packet(s) received
 0.00% packet loss
 round-trip min/avg/max = 0/6/16 ms

```

(8) PC-10-1 和 RTD 上的 loopback 0 不能 ping 通:

The screenshot shows a terminal window titled "PC-10-1" with tabs for "基础配置", "命令行", "组播", "UDP发包工具", and "串口". The "命令行" tab is active, displaying the following output:

```

PC>ping 192.168.200.100

Ping 192.168.200.100: 32 data bytes, Press Ctrl_C to break
Request timeout!
Request timeout!
Request timeout!
Request timeout!
Request timeout!

--- 192.168.200.100 ping statistics ---
 5 packet(s) transmitted
 0 packet(s) received
100.00% packet loss

```

Q4-1.9~4-1.15

(9) RTA 的 IP 路由表如下:

```

[RTA]display IP routing-table
Route Flags: R - relay, D - download to fib
-----
Routing Tables: Public
  Destinations : 11          Routes : 11

Destination/Mask    Proto    Pre  Cost    Flags NextHop        Interface
-----
127.0.0.0/8         Direct   0     0        D  127.0.0.1        InLoopBack0
127.0.0.1/32        Direct   0     0        D  127.0.0.1        InLoopBack0
127.255.255.255/32  Direct   0     0        D  127.0.0.1        InLoopBack0
172.16.101.0/24     Direct   0     0        D  172.16.101.1     GigabitEthernet
0/0/0
172.16.101.1/32     Direct   0     0        D  127.0.0.1        GigabitEthernet
0/0/0
172.16.101.255/32   Direct   0     0        D  127.0.0.1        GigabitEthernet
0/0/0
192.168.10.0/24     Direct   0     0        D  192.168.10.1     GigabitEthernet
0/0/2
192.168.10.1/32     Direct   0     0        D  127.0.0.1        GigabitEthernet
0/0/2
192.168.10.255/32   Direct   0     0        D  127.0.0.1        GigabitEthernet
0/0/2
192.168.30.0/24     Static   60    0        RD  172.16.101.2     GigabitEthernet
0/0/0
255.255.255.255/32  Direct   0     0        D  127.0.0.1        InLoopBack0

```

(10) RTB 的 IP 路由表如下：

```
[RTB]display IP routing-table
Route Flags: R - relay, D - download to fib
-----
```

Routing Tables: Public						
Destinations : 12			Routes : 12			
Destination/Mask	Proto	Pre	Cost	Flags	NextHop	Interface
127.0.0.0/8	Direct	0	0	D	127.0.0.1	InLoopBack0
127.0.0.1/32	Direct	0	0	D	127.0.0.1	InLoopBack0
127.255.255.255/32	Direct	0	0	D	127.0.0.1	InLoopBack0
172.16.101.0/24	Direct	0	0	D	172.16.101.2	GigabitEthernet
0/0/0						
172.16.101.2/32	Direct	0	0	D	127.0.0.1	GigabitEthernet
0/0/0						
172.16.101.255/32	Direct	0	0	D	127.0.0.1	GigabitEthernet
0/0/0						
172.16.102.0/24	Direct	0	0	D	172.16.102.1	GigabitEthernet
0/0/1						
172.16.102.1/32	Direct	0	0	D	127.0.0.1	GigabitEthernet
0/0/1						
172.16.102.255/32	Direct	0	0	D	127.0.0.1	GigabitEthernet
0/0/1						
192.168.10.0/24	Static	60	0	RD	172.16.101.1	GigabitEthernet
0/0/0						
192.168.30.0/24	Static	60	0	RD	172.16.102.2	GigabitEthernet
0/0/1						
255.255.255.255/32	Direct	0	0	D	127.0.0.1	InLoopBack0

(11) RTC 的 IP 路由表如下：

```
[RTC]display IP routing-table
Route Flags: R - relay, D - download to fib
-----
```

Routing Tables: Public						
Destinations : 12			Routes : 12			
Destination/Mask	Proto	Pre	Cost	Flags	NextHop	Interface
127.0.0.0/8	Direct	0	0	D	127.0.0.1	InLoopBack0
127.0.0.1/32	Direct	0	0	D	127.0.0.1	InLoopBack0
127.255.255.255/32	Direct	0	0	D	127.0.0.1	InLoopBack0
172.16.102.0/24	Direct	0	0	D	172.16.102.2	GigabitEthernet
0/0/1						
172.16.102.2/32	Direct	0	0	D	127.0.0.1	GigabitEthernet
0/0/1						
172.16.102.255/32	Direct	0	0	D	127.0.0.1	GigabitEthernet
0/0/1						
172.16.103.0/24	Direct	0	0	D	172.16.103.2	GigabitEthernet
0/0/0						
172.16.103.2/32	Direct	0	0	D	127.0.0.1	GigabitEthernet
0/0/0						
172.16.103.255/32	Direct	0	0	D	127.0.0.1	GigabitEthernet
0/0/0						
192.168.10.0/24	Static	60	0	RD	172.16.102.1	GigabitEthernet
0/0/1						
192.168.30.0/24	Static	60	0	RD	172.16.103.1	GigabitEthernet
0/0/0						
255.255.255.255/32	Direct	0	0	D	127.0.0.1	InLoopBack0

(12) RTD 的 IP 路由表如下：

```
[RTD]display IP routing-table
Route Flags: R - relay, D - download to fib
-----
```

Routing Tables: Public						
Destinations : 14			Routes : 14			
Destination/Mask	Proto	Pre	Cost	Flags	NextHop	Interface
127.0.0.0/8	Direct	0	0	D	127.0.0.1	InLoopBack0
127.0.0.1/32	Direct	0	0	D	127.0.0.1	InLoopBack0
127.255.255.255/32	Direct	0	0	D	127.0.0.1	InLoopBack0
172.16.103.0/24	Direct	0	0	D	172.16.103.1	GigabitEthernet
0/0/0						
172.16.103.1/32	Direct	0	0	D	127.0.0.1	GigabitEthernet
0/0/0						
172.16.103.255/32	Direct	0	0	D	127.0.0.1	GigabitEthernet
0/0/0						
192.168.10.0/24	Static	60	0	RD	172.16.103.2	GigabitEthernet
0/0/0						
192.168.30.0/24	Direct	0	0	D	192.168.30.1	GigabitEthernet
0/0/2						
192.168.30.1/32	Direct	0	0	D	127.0.0.1	GigabitEthernet
0/0/2						
192.168.30.255/32	Direct	0	0	D	127.0.0.1	GigabitEthernet
0/0/2						
192.168.200.0/24	Direct	0	0	D	192.168.200.100	LoopBack0
192.168.200.100/32	Direct	0	0	D	127.0.0.1	LoopBack0
192.168.200.255/32	Direct	0	0	D	127.0.0.1	LoopBack0
255.255.255.255/32	Direct	0	0	D	127.0.0.1	InLoopBack0

(13) PC-10-1 和 PC-30-1 可以 ping 通:

```
PC>ping 192.168.30.11

Ping 192.168.30.11: 32 data bytes, Press Ctrl_C to break
From 192.168.30.11: bytes=32 seq=1 ttl=124 time=16 ms
From 192.168.30.11: bytes=32 seq=2 ttl=124 time=31 ms
From 192.168.30.11: bytes=32 seq=3 ttl=124 time=31 ms
From 192.168.30.11: bytes=32 seq=4 ttl=124 time=32 ms
From 192.168.30.11: bytes=32 seq=5 ttl=124 time=31 ms

--- 192.168.30.11 ping statistics ---
 5 packet(s) transmitted
 5 packet(s) received
 0.00% packet loss
 round-trip min/avg/max = 16/28/32 ms
```

(14) PC-10-1 和 RTD 上的 loopback 0 不能 ping 通, 因为各路由器均未配置到 RTD 的 Loopback 0 (192.168.200.0/24) 的路由。

```
PC>ping 192.168.200.100

Ping 192.168.200.100: 32 data bytes, Press Ctrl_C to break
Request timeout!
Request timeout!
Request timeout!
Request timeout!
Request timeout!

--- 192.168.200.100 ping statistics ---
 5 packet(s) transmitted
 0 packet(s) received
100.00% packet loss
```

(15) PC-10-1 上 “tracert 192.168.30.11” 命令执行结果如下:

```
PC>tracert 192.168.30.11

tracert to 192.168.30.11, 8 hops max
(ICMP), press Ctrl+C to stop
 1  192.168.10.1      31 ms  <1 ms  16 ms
 2  172.16.101.2     31 ms  16 ms  15 ms
 3  172.16.102.2     31 ms  32 ms  15 ms
 4  172.16.103.1     16 ms  31 ms  31 ms
 5  192.168.30.11    32 ms  31 ms  16 ms
```

Q4-1.16~4-1.20

(16) RTA 的 IP 路由表如下:

```
[RTA]display IP routing-table
Route Flags: R - relay, D - download to fib
-----
Routing Tables: Public
Destinations : 12      Routes : 12

Destination/Mask    Proto Pre  Cost   Flags NextHop         Interface
-----
0.0.0.0/0           Static 60   0       RD  172.16.101.2      GigabitEthernet
0/0/0
127.0.0.0/8         Direct 0     0       D   127.0.0.1         InLoopBack0
127.0.0.1/32        Direct 0     0       D   127.0.0.1         InLoopBack0
127.255.255.255/32   Direct 0     0       D   127.0.0.1         InLoopBack0
172.16.101.0/24      Direct 0     0       D   172.16.101.1      GigabitEthernet
0/0/0
172.16.101.1/32     Direct 0     0       D   127.0.0.1         GigabitEthernet
0/0/0
172.16.101.255/32   Direct 0     0       D   127.0.0.1         GigabitEthernet
0/0/0
192.168.10.0/24     Direct 0     0       D   192.168.10.1      GigabitEthernet
0/0/2
192.168.10.1/32     Direct 0     0       D   127.0.0.1         GigabitEthernet
0/0/2
192.168.10.255/32   Direct 0     0       D   127.0.0.1         GigabitEthernet
0/0/2
192.168.30.0/24     Static 60   0       RD  172.16.101.2      GigabitEthernet
0/0/0
255.255.255.255/32   Direct 0     0       D   127.0.0.1         InLoopBack0
```

(17) RTB 的 IP 路由表如下:

```
[RTB]display IP routing-table
Route Flags: R - relay, D - download to fib
-----
Routing Tables: Public
Destinations : 13      Routes : 13

Destination/Mask    Proto Pre  Cost   Flags NextHop         Interface
-----
0.0.0.0/0           Static 60   0       RD  172.16.102.2      GigabitEthernet
0/0/1
127.0.0.0/8         Direct 0     0       D   127.0.0.1         InLoopBack0
127.0.0.1/32        Direct 0     0       D   127.0.0.1         InLoopBack0
127.255.255.255/32   Direct 0     0       D   127.0.0.1         InLoopBack0
172.16.101.0/24      Direct 0     0       D   172.16.101.2      GigabitEthernet
0/0/0
172.16.101.2/32     Direct 0     0       D   127.0.0.1         GigabitEthernet
0/0/0
172.16.101.255/32   Direct 0     0       D   127.0.0.1         GigabitEthernet
0/0/0
172.16.102.0/24     Direct 0     0       D   172.16.102.1      GigabitEthernet
0/0/1
172.16.102.1/32     Direct 0     0       D   127.0.0.1         GigabitEthernet
0/0/1
172.16.102.255/32   Direct 0     0       D   127.0.0.1         GigabitEthernet
0/0/1
192.168.10.0/24     Static 60   0       RD  172.16.101.1      GigabitEthernet
0/0/0
192.168.30.0/24     Static 60   0       RD  172.16.102.2      GigabitEthernet
0/0/1
255.255.255.255/32   Direct 0     0       D   127.0.0.1         InLoopBack0
```

(18) RTC 的 IP 路由表如下:

```
[RTC]display IP routing-table
Route Flags: R - relay, D - download to fib
-----
Routing Tables: Public
Destinations : 13      Routes : 13

Destination/Mask    Proto Pre  Cost   Flags NextHop         Interface
-----
0.0.0.0/0           Static 60   0       RD  172.16.103.1      GigabitEthernet
0/0/0
127.0.0.0/8         Direct 0     0       D   127.0.0.1         InLoopBack0
127.0.0.1/32        Direct 0     0       D   127.0.0.1         InLoopBack0
127.255.255.255/32   Direct 0     0       D   127.0.0.1         InLoopBack0
172.16.102.0/24      Direct 0     0       D   172.16.102.2      GigabitEthernet
0/0/1
172.16.102.2/32     Direct 0     0       D   127.0.0.1         GigabitEthernet
0/0/1
172.16.102.255/32   Direct 0     0       D   127.0.0.1         GigabitEthernet
0/0/1
172.16.103.0/24     Direct 0     0       D   172.16.103.2      GigabitEthernet
0/0/0
172.16.103.2/32     Direct 0     0       D   127.0.0.1         GigabitEthernet
0/0/0
172.16.103.255/32   Direct 0     0       D   127.0.0.1         GigabitEthernet
0/0/0
192.168.10.0/24     Static 60   0       RD  172.16.102.1      GigabitEthernet
0/0/1
192.168.30.0/24     Static 60   0       RD  172.16.103.1      GigabitEthernet
0/0/0
255.255.255.255/32   Direct 0     0       D   127.0.0.1         InLoopBack0
```

(19) RTD 的 IP 路由表如下:

```
[RTD]ip route-static 0.0.0.0 0.0.0.0 172.16.103.2
[RTD]display IP routing-table
Route Flags: R - relay, D - download to fib
-----
Routing Tables: Public
Destinations : 14      Routes : 14

Destination/Mask    Proto   Pre  Cost   Flags NextHop         Interface
-----
0/0/0              0.0.0.0/0   Static 60    0      RD  172.16.103.2      GigabitEthernet
127.0.0.0/8        127.0.0.0/8   Direct 0     0      D   127.0.0.1         InLoopBack0
127.0.0.1/32       127.0.0.1/32   Direct 0     0      D   127.0.0.1         InLoopBack0
127.255.255.255/32 127.255.255.255/32 Direct 0     0      D   127.0.0.1         InLoopBack0
172.16.103.0/24    172.16.103.0/24 Direct 0     0      D   172.16.103.1      GigabitEthernet
172.16.103.1/32    172.16.103.1/32 Direct 0     0      D   127.0.0.1         GigabitEthernet
172.16.103.255/32 172.16.103.255/32 Direct 0     0      D   127.0.0.1         GigabitEthernet
192.168.30.0/24    192.168.30.0/24 Direct 0     0      D   192.168.30.1      GigabitEthernet
192.168.30.1/32    192.168.30.1/32 Direct 0     0      D   127.0.0.1         GigabitEthernet
192.168.30.255/32 192.168.30.255/32 Direct 0     0      D   127.0.0.1         GigabitEthernet
192.168.200.0/24   192.168.200.0/24 Direct 0     0      D   192.168.200.100   LoopBack0
192.168.200.100/32 192.168.200.100/32 Direct 0     0      D   127.0.0.1         LoopBack0
192.168.200.255/32 192.168.200.255/32 Direct 0     0      D   127.0.0.1         LoopBack0
255.255.255.255/32 255.255.255.255/32 Direct 0     0      D   127.0.0.1         InLoopBack0
```

(20) PC-10-1 和 RTD 上 loopback0 可以 ping 通，因为现在双向路由均完整，发送与接收线路上默认路由与静态路由协同工作，可以确保数据包正确转发。

PC-10-1

基础配置

命令行

组播

UDP发包工具

串口

```
PC>ping 192.168.200.100

Ping 192.168.200.100: 32 data bytes, Press Ctrl_C to break
From 192.168.200.100: bytes=32 seq=1 ttl=252 time=16 ms
From 192.168.200.100: bytes=32 seq=2 ttl=252 time=31 ms
From 192.168.200.100: bytes=32 seq=3 ttl=252 time=31 ms
From 192.168.200.100: bytes=32 seq=4 ttl=252 time=31 ms
From 192.168.200.100: bytes=32 seq=5 ttl=252 time=31 ms

--- 192.168.200.100 ping statistics ---
 5 packet(s) transmitted
 5 packet(s) received
 0.00% packet loss
 round-trip min/avg/max = 16/28/31 ms
```